InfoGate: Information-Bottleneck-Guided Adaptive Cross-Attention for Multimodal Sentiment Analysis

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Abstract

Multimodal Sentiment Analysis (MSA) integrates language, acoustic and visual signals to estimate utterance-level sentiment intensity. State-of-the-art systems combine cross-modal attention with information-bottleneck (IB) compression, but they (i) weight every auxiliary token equally, regardless of how informative it is, (ii) inject auxiliary cues with a fixed strength even when the auxiliary signal contradicts the primary modality, and (iii) commit to a single language-centric primary modality before observing the sample. We present InfoGate, a framework that uses a single signal – the per-token confidence $\sigma(-\log\sigma^2)$ extracted from a variational IB encoder – to modulate three different stages of multimodal fusion: token-level attention reweighting, sample-level alignment-modulated injection, and global per-sample primary-modality routing. InfoGate is trained end-to-end with a task loss, token- and label-IB losses, and a stage-2 routing-IB loss that supervises the modality selector with per-modality task error. With Bayesian hyper-parameter search on three benchmarks, InfoGate reaches state-of-the-art MAE on CMU-MOSI (0.5923), CMU-MOSEI (0.4941) and CH-SIMS v2 (0.3138), while also offering a competing trade-off with 88.85% Acc-2 / 88.83% F1 on CMU-MOSI.

1 Introduction

Multimodal Sentiment Analysis (Zadeh et al., 2016, 2018c; Liu et al., 2022a) predicts the sentiment intensity of an utterance from three loosely aligned streams: language, acoustic prosody and facial dynamics. The dominant design pattern is to use a strong text encoder (BERT-style) as the primary backbone and inject acoustic / visual context through cross-modal attention or low-rank fusion (Rahman et al., 2020; Tsai et al., 2019a; Yu et al., 2021; Han et al., 2021). Recent work adds an information-bottleneck (IB) regulariser to compress each modality into a task-relevant subspace (Anonymous, 2024f; Wu et al., 2023; Anonymous, 2023, 2026), which improves the state-of-the-art MAE on CMU-MOSI and CMU-MOSEI.

Three weaknesses persist across this line of work: **(W1)** Cross-modal attention treats every auxiliary token as equally informative. A whispered *"yeah"* and a screamed *"yeah"* have identical text content but very different acoustic information; current key/value gating does not see this distinction. **(W2)** The injection strength of auxiliary tokens into the primary stream is fixed regardless of *cross-modal consistency*. A smiling face during a sarcastic verbal utterance contradicts the language signal, but is added to the primary representation with the same weight as a corroborating face. **(W3)** Most architectures pre-commit to a fixed primary modality, usually text. Recent dynamic-routing approaches such as MODS (Wang et al., 2026) introduce a learned modality selector, but the selector is trained from the task gradient alone and tends to collapse onto the easy text-only prior.

We argue that all three weaknesses share a single root cause: the absence of a calibrated, sample-wise notion of *information quality* that can be propagated through the fusion pipeline. We provide one. The variational IB encoder used by ITHP and CyIN already produces a per-token log-variance $\log\sigma^{2,m}$, and we observe that the corresponding confidence $\mathbf{c}^m = \sigma(-\log\sigma^{2,m})$ is a natural, end-to-end-learnable measure of how task-relevant a token is. We use it once to modulate *everything* downstream.

Concretely, our model InfoGate (Figure 1) uses $\mathbf{c}^m$ at three different scales:

- **Token level** – IB-Guided Cross-Attention biases scores by $\gamma\log\bar{c}^a$ and gates values by $\bar{c}^a$, so noisy auxiliary positions are simultaneously down-weighted in attention and in contribution (§3.3).
- **Sample level** – Alignment-Modulated Injec-

tion scales each cross-attention output by the cosine similarity of the primary and auxiliary bottleneck representations, lower-bounded at 0.3; the injection therefore contracts when modalities disagree (§3.4).

- **Global level** – a Gumbel-softmax modality selector picks one of the three modalities as the primary stream per sample, supervised in stage 2 by a routing-IB loss whose teacher signal is the per-modality task error in the bottleneck space (§3.6).

This single mechanism resolves **W1–W3** simultaneously and is fully differentiable through a single $\log \sigma^2$ pathway.

Our contributions are: (i) the *IB-confidence-as-modulation* principle that unifies token, sample and global filtering through one back-propagatable signal; (ii) the IB-Guided Cross-Attention mechanism with score bias and value gate; (iii) the Alignment-Modulated Injection rule for sample-level cross-modal consistency control; (iv) a stage-2 routing-IB loss that supervises the modality selector with per-modality task error; and (v) state-of-the-art results on CMU-MOSI (MAE 0.5923) and CMU-MOSEI (MAE 0.4941) under careful Optuna search, with a competitive trade-off on CH-SIMS v2. The paper is organised as follows. §2 reviews multimodal sentiment fusion and IB-based methods. §3 formalises InfoGate. §4 reports the main results, ablations and hyper-parameter search. §5 analyses routing behaviour and the MAE-vs-accuracy trade-off. §6 concludes; limitations and ethics statements follow.

2 Related Work

Multimodal sentiment fusion. Early work on CMU-MOSI and CMU-MOSEI aligned modality-specific recurrent encoders through tensor / outer-product fusion (TFN (Zadeh et al., 2017), LMF (Liu et al., 2018), MFN (Zadeh et al., 2018a), Graph-MFN (Zadeh et al., 2018b)). The transformer era replaced these with cross-modal attention (Tsai et al., 2019a) and BERT-grounded fusion (MAG-BERT (Rahman et al., 2020), MISA (Hazarika et al., 2020), Self-MM (Yu et al., 2021), MMIM (Han et al., 2021)). A second line of work disentangles modality-shared and modality-specific subspaces (MFM (Tsai et al., 2019b), DMD (Li et al., 2023), EMOE (Anonymous, 2024d)); a third focuses on cross-modal alignment through contrastive or denoising objectives (Wu et al., 2023; Anonymous, 2024h).

Information bottleneck for multimodal learning. The IB principle (Tishby et al., 1999; Tishby and Zaslavsky, 2015) compresses input X into a bottleneck B by minimising $I(X;B)$ while maximising $I(B;Y)$. Its variational form (VIB) (Alemi et al., 2017; Kingma and Welling, 2014) makes the objective tractable for deep networks. ITHP (Anonymous, 2024f) introduces a hierarchical IB chain for multimodal perception, while CyIN cyclically translates between modality bottlenecks; CaMIB (Anonymous, 2026) adds causal-aware ingredients to the same recipe. C-MIB (Anonymous, 2023) and DLF (Anonymous, 2024c) regularise cross-modal mutual information. InfoGate differs from this family in two ways. First, while existing IB methods use $\log \sigma^2$ only inside the KL term of the VIB objective, we expose it as a *first-class modulation signal* for downstream attention and gating. Second, IB methods that perform translation (CyIN, ITHP) require auxiliary cross-modal decoders even at inference; InfoGate does not perform any cross-modal translation and therefore has a strictly smaller inference-time graph.

Routing and dynamic primary selection. MODS (Wang et al., 2026) is the closest predecessor. It introduces a Modality Selector that uses Gumbel-softmax to pick a per-sample primary modality, and an equal-weight cross-attention fusion (PCCA). InfoGate reuses the MSelector backbone, but adds the IB-guided modulation of §3.3–3.5, and supervises the selector in stage 2 with a routing-IB loss that uses the per-modality task error rather than a fixed text prior. DEVA (Anonymous, 2024b), KuDA (Anonymous, 2024g), MOAC (Anonymous, 2025) and TMSON (Anonymous, 2024i) are concurrent dynamic-routing or adaptive-fusion methods compared in Table 1; none of them couples routing to a per-token IB confidence. Adaptive Information Gates (Eq. 10) are also reminiscent of the gated fusion in MAG-BERT and DBF (Wu et al., 2023), but condition on the IB-compressed bottleneck rather than on raw features.

3 Method

3.1 Overview

Let an utterance be represented by three temporally aligned token sequences $X^t \in \mathbb{R}^{T \times d_t}$ (text, last hidden states of DeBERTa-v3-base (He et al., 2023)), $X^a \in \mathbb{R}^{T \times d_a}$ (acoustic, COVAREP (Degottex et al., 2014)) and $X^v \in \mathbb{R}^{T \times d_v}$ (visual, OpenFace 2.0 (Baltrusaitis et al., 2018)). The goal is to predict a continuous sentiment score $y \in \mathbb{R}$.

InfoGate is built around a single design principle: the *IB-derived per-token confidence* $\mathbf{c} = \sigma(-\log \sigma^2) \in [0,1]^{T \times d_B}$ is a universal control signal that can modulate fusion at three different scales (Figure 1): **(i) token level** – biasing cross-attention scores and gating values (§3.3); **(ii) sample level** – modulating the magnitude of cross-modal injection by the cosine similarity of the bottleneck representations (§3.4); **(iii) global level** – supervising a Gumbel–softmax modality selector through a stage-2 routing-IB loss (§3.6).

3.2 Per-Modality Projection and IB Encoder

Each modality is first projected into a shared d_u-dimensional space with $\mathrm{Proj}^m(X^m)$ realised as a linear layer plus ReLU (text), or LayerNorm-Linear-ReLU (acoustic / visual; the LayerNorm stabilises the very different feature statistics of the raw COVAREP / OpenFace streams). Each projected sequence $F^m \in \mathbb{R}^{T \times d_u}$ is then fed to a variational IB encoder IBE^m that returns mean μ^m and log-variance $\log \sigma^{2,m}$, from which a stochastic bottleneck representation is sampled with the standard reparameterization trick (Kingma and Welling, 2014; Alemi et al., 2017):

$$(\mu^m, \log \sigma^{2,m}) = \mathrm{IBE}^m(F^m), \tag{1}$$

$$\mathbf{B}^m = \mu^m + \boldsymbol{\epsilon} \odot \exp(\tfrac{1}{2} \log \sigma^{2,m}), \tag{2}$$

with $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$ during training and $\mathbf{B}^m = \mu^m$ at evaluation. The IB principle (Tishby et al., 1999; Tishby and Zaslavsky, 2015; Alemi et al., 2017) encourages the bottleneck to retain only task-relevant information, making $\log \sigma^{2,m}$ a natural proxy for per-token uncertainty. We therefore define a per-token, per-dimension *confidence map*

$$\mathbf{c}^m = \sigma\big(-\log \sigma^{2,m}\big) \in [0,1]^{T \times d_B}, \tag{3}$$

which is reused at every later modulation stage.

To make the bottleneck informative we attach an intra-modal decoder Dec^m for each modality and minimise the token-level IB objective

$$\mathcal{L}_{\text{tib}} = \sum_{m \in \{t,a,v\}} \mathrm{KL}\big(\mu^m, \sigma^{2,m}\big) + \beta_{\text{ib}}\, \mathrm{MSE}\big(\mathrm{Dec}^m(\mathbf{B}^m), F^m\big), \tag{4}$$

where $\mathrm{KL}(\mu, \sigma^2) = -\frac{1}{2}\sum_i \big(1 + \log \sigma_i^2 - \mu_i^2 - \sigma_i^2\big)$ is the closed-form KL against $\mathcal{N}(0, \mathbf{I})$, masked over valid tokens. A label-level IB term anchors the bottleneck to the sentiment label through a per-modality linear head LP^m:

$$\mathcal{L}_{\text{lib}} = \frac{1}{3} \sum_m \Big[\mathrm{KL}(\mu_{\bar{t}}^m, \sigma_{\bar{t}}^{2,m}) + \beta_{\text{ib}} \left|\mathrm{LP}^m(\mathbf{B}_{\bar{t}}^m) - y\right|\Big], \tag{5}$$

where $\bar{t}$ denotes mean-pooling over valid tokens.

3.3 IB-Guided Cross-Attention

Cross-attention from auxiliary modality a to primary modality p is the standard scaled dot-product mechanism with two confidence-driven modifications. For each head, queries are projected from $\mathbf{B}^p$ and keys/values from $\mathbf{B}^a$ as Q, K, V. Let $\bar{c}^a \in \mathbb{R}^T$ be the mean of $\mathbf{c}^a$ over the bottleneck dimension. We modify the attention as

$$\mathbf{S} = \tfrac{1}{\sqrt{d_h}} QK^\top + \gamma \log(\bar{c}^a + \varepsilon), \tag{6}$$

$$\mathbf{V}_{\text{eff}} = V \odot \bar{c}^a, \tag{7}$$

$$\mathrm{Attn}(Q, K, V, \mathbf{c}^a) = \mathrm{softmax}(\mathbf{S} + \mathbf{M})\mathbf{V}_{\text{eff}}, \tag{8}$$

where γ is a learnable scalar (initialised to 1), $\varepsilon = 10^{-6}$ and $\mathbf{M}$ is the standard padding mask. The score bias (6) discourages attention from being placed on uncertain key positions, and the value gate (7) shrinks the contribution of those positions even when the softmax nevertheless attends to them. Figure 2 visualises the data path. This design is fully back-propagatable through $\log \sigma^{2,a}$, so the IB encoder learns to suppress its own outputs whenever they are not useful for the downstream prediction.

3.4 Alignment-Modulated Injection

Auxiliary tokens may be uninformative not only in absolute terms (captured by $\mathbf{c}^a$) but also *relative* to the primary modality – e.g. a smiling face during a sarcastic verbal utterance. We model this with a sample-level alignment scalar

$$\alpha_i^{p,a} = \max\Big(0.3,\ \tfrac{1}{2}\big(\cos(\bar{\mathbf{B}}_i^p, \bar{\mathbf{B}}_i^a) + 1\big)\Big) \in [0.3, 1], \tag{9}$$

where $\bar{\mathbf{B}}_i^m$ is the masked-mean of $\mathbf{B}^m$ for sample i. The lower bound at 0.3 prevents the auxiliary path from being closed entirely, which we found to destabilise training when the bottleneck representations have not yet converged. $\alpha_i^{p,a}$ multiplies the gated cross-attention output before it is added to the

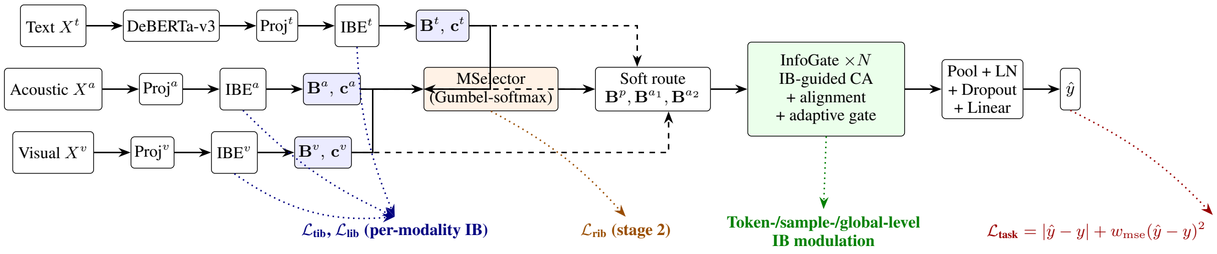

Figure 1: Overview of InfoGate. Each modality is projected, encoded by a variational IB encoder that exposes a per-token confidence $\mathbf{c}^m = \sigma(-\log \sigma^{2,m})$, and routed by a Gumbel-softmax modality selector. The InfoGate stack (§3.3–3.5) uses the same confidence signal at three different scales (token, sample, global) to modulate cross-attention, alignment-aware injection and adaptive gating before a primary-only regression head.

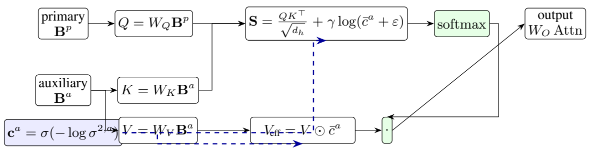

Figure 2: IB-Guided Cross-Attention for one head. Per-token confidence $\mathbf{c}^a$ from the IB encoder of the auxiliary modality is used in two places: (i) as a log-bias on attention scores (suppressing uncertain keys) and (ii) as a multiplicative gate on values (suppressing uncertain contributions). Both effects are back-propagatable through $\log \sigma^{2,a}$.

primary stream (§3.5), giving a sample-adaptive injection weight that contracts to 0.3 when the two modalities disagree.

3.5 Adaptive Information Gate

The vanilla way to mix a cross-attention output with the primary stream is a residual addition. We instead learn a per-sample, per-dimension gate

$$\mathbf{g}^a = \sigma\big(\mathbf{W}_2\,\text{ReLU}(\mathbf{W}_1[\mathbf{B}^p \,\|\, \text{CA}^a])\big), \qquad (10)$$

where CA^a is the IB-guided cross-attention output for auxiliary a and $[\cdot\|\cdot]$ denotes concatenation along the feature axis. The fused primary representation in one InfoGate layer is then

$$\mathbf{B}^p \leftarrow \mathbf{B}^p + \text{SA}^p + \sum_{a \in \{a_1, a_2\}} \alpha^{p,a} \left(\mathbf{g}^a \odot \text{CA}^a\right), \qquad (11)$$

followed by LayerNorm and a position-wise feed-forward block. Stacking N such layers yields the InfoGate fusion stack; in all but the last layer we additionally back-propagate primary information into the two auxiliary streams to maintain residual alignment.

3.6 Modality Selector and Routing-IB Loss

For each sample we want to choose which of the three modalities should play the role of the primary stream. Modality Selector (MSelector) computes a single score per modality with adaptive attention pooling

$$h^m = \sum_{t=1}^{T} \text{softmax}_t\big(\mathbf{w}^{m\top}\mathbf{B}^m / \sqrt{d_B}\big)\, \mathbf{B}^m_t, \qquad (12)$$

followed by a 2-layer MLP that produces logits $\boldsymbol{\ell} = \text{MLP}([h^a; h^t; h^v]) \in \mathbb{R}^3$. At training time we draw a hard one-hot vector with a Gumbel–softmax (Jang et al., 2017; Maddison et al., 2017) $\mathbf{o} = \text{GumbelSoftmax}(\boldsymbol{\ell}, \tau)$, kept differentiable through the straight-through estimator, while the soft weights $\mathbf{w} = \text{softmax}(\boldsymbol{\ell}/\tau)$ scale the auxiliary streams; τ is annealed linearly from τ_{start} to τ_{end} over training. At inference we use $\arg\max(\boldsymbol{\ell})$.

To prevent MSelector from collapsing onto a fixed text-centric prior that simply mimics MAG-BERT (Rahman et al., 2020) or Self-MM (Yu et al., 2021), we add a stage-2 routing-IB loss that supervises the soft weights with a per-sample modality-quality target. Specifically, we use the per-modality label-prediction errors as a teacher signal

$$\mathbf{q}_i = \text{softmax}\big(-|\text{LP}^m(\mathbf{B}^m_{\bar{t}}) - y_i|/\tau_q\big), \qquad (13)$$

and minimise the per-sample KL divergence on the subset of samples for which the modalities actually differ in quality ($\text{range}(\text{errors}) > 0.5$):

$$\mathcal{L}_{\text{rib}} = \mathbb{E}_{i \in \mathcal{D}}\,\text{KL}(\mathbf{w}_i \,\|\, \mathbf{q}_i) + \lambda_b \left(\log 3 - H(\bar{\mathbf{w}})\right), \qquad (14)$$

where $\bar{\mathbf{w}}$ is the batch-mean weight vector and the second term is a mild entropy regulariser ($\lambda_b = 0.0$ in our final configurations). $\mathcal{L}_{\text{rib}}$ is only computed in stage 2 (after stage1_epochs warm-up) so that the per-modality label predictors LP^m are already meaningful before they are used as teachers.

3.7 Training Objective

Putting it all together, InfoGate optimises

$$\mathcal{L} = \mathcal{L}_{\text{task}} + \alpha_{\text{ib}}\,(\mathcal{L}_{\text{tib}}+\mathcal{L}_{\text{lib}}) + \lambda_{\text{rib}}\cdot\mathbb{1}_{[\text{stage}=2]}\,\mathcal{L}_{\text{rib}}, \tag{15}$$

with the regression task loss

$$\mathcal{L}_{\text{task}} = |\hat{y} - y| + w_{\text{mse}}\,(\hat{y} - y)^2, \tag{16}$$

where $\hat{y}$ is the scalar output of a LayerNorm–Dropout–Linear head applied to the adaptively pooled primary bottleneck. The weight w_{mse} explicitly trades MAE for Acc-2 / F1 classification quality (§4.3); we report two configurations on CMU-MOSI that sit at different points of this trade-off. A two-stage schedule defines stage $= 1$ if the current epoch is below stage1_epochs and stage $= 2$ thereafter; only stage 2 activates the routing supervision in Equation (14).

We use AdamW (Loshchilov and Hutter, 2019; Kingma and Ba, 2015) with two parameter groups (a small learning rate for DeBERTa, a 10–40× larger one for the InfoGate parameters), linear warm-up over a warmup_proportion fraction of total steps, and an exponential moving average of the model weights for evaluation after ema_start_epoch. Hyper-parameter search uses Optuna (Akiba et al., 2019) (§4.1).

4 Experiments

4.1 Datasets and Evaluation

We evaluate InfoGate on three standard MSA benchmarks. **CMU-MOSI** (Zadeh et al., 2016) contains 1,281 / 229 / 685 English monologue utterances (train / dev / test) annotated with sentiment intensity in $[-3, 3]$. **CMU-MOSEI** (Zadeh et al., 2018c) extends MOSI to 16,265 / 1,869 / 4,643 utterances drawn from 1,000 YouTube speakers. **CH-SIMS v2** (Liu et al., 2022a) contains 2,722 / 647 / 1,034 Mandarin video clips with sentiment intensity in $[-1, 1]$, designed to make acoustic and visual cues genuinely informative. For all three datasets we use the official feature releases: COVAREP (Degottex et al., 2014) for audio and OpenFace 2.0 (Baltrusaitis et al., 2018) for vision, with input dimensions $(d_a, d_v) = (74, 47)$ for MOSI, $(74, 35)$ for MOSEI and $(25, 177)$ for SIMSv2. We report Acc-2 (positive vs. negative), F1, MAE, Pearson Corr. and the multi-class accuracies that are conventional for each benchmark (Acc-7 for MOSI/MOSEI; Acc-3 / Acc-5 for SIMSv2). For MOSI/MOSEI we use the "negative vs. non-negative" binary protocol, which excludes neutral samples, following Tsai et al. (2019a); Yu et al. (2021).

4.2 Implementation Details

We use DeBERTa-v3-base (He et al., 2023) as the text backbone for MOSI/MOSEI and BERT-base-Chinese (Devlin et al., 2019) for SIMSv2. Each modality is projected to a unified $d_u = 256$ space (§3.2), and the IB encoder hidden width is also 256. We fix the bottleneck width to $d_B = 96$ for MOSI / MOSEI and $d_B = 128$ for SIMSv2 after Optuna search. The InfoGate stack uses 4 heads and $N \in \{3, 4, 5\}$ layers (per-dataset best, see Appendix A). We optimise with AdamW (Loshchilov and Hutter, 2019) using two parameter groups: a small learning rate ($\sim 10^{-5}$) for DeBERTa / BERT and a larger one ($\sim 10^{-4}$) for InfoGate parameters; both are scheduled with linear warm-up over a fraction $\text{warmup\_proportion} \in [0.05, 0.2]$ of total steps. An exponential moving average (EMA) of model weights is taken from the 5-th epoch onward and used for evaluation. The two-stage schedule activates $\mathcal{L}_{\text{rib}}$ from epoch stage1_epochs onward, where stage1_epochs is itself a tunable hyper-parameter.

Hyper-parameter search. We use the Optuna framework (Akiba et al., 2019) with a two-stage search per dataset: a stage-1 *random* sampler explores the global space (Tier-1 = 9 knobs: batch / accumulation, n_{epochs}, DeBERTa LR, InfoGate LR, β_{ib}, N, d_B, w_{mse}, dropout); a stage-2 TPE sampler *locally* re-samples around the top-8 stage-1 trials and adds Tier-2 (α_{ib}, stage1_epochs, warmup_proportion, weight decay, ema_decay) and, optionally, Tier-3 (selector temperatures, Gumbel τ schedule, head count, d_u, ema_start_epoch). For each dataset we ran 35–200 trials per study, selected the best checkpoint by dev-MAE with a tie-break on Acc-2 / F1 / Acc-k / Corr, and finally report the test metrics of the selected checkpoint.

4.3 Main Results

Tables 1 and 2 compare InfoGate with 14 recent baselines on MOSI / MOSEI and 13 baselines on CH-SIMS v2; baseline numbers are taken from the corresponding papers. Two InfoGate configurations are reported on MOSI: the **MAE-tuned** configuration that minimises dev-MAE (our default), and an **Acc-tuned** configuration found by the same search with $w_{\text{mse}} = 2.87$ (vs. 1.25 for MAE-tuned), exposing the classification-vs.-regression trade-off discussed in §3.7. On the two English benchmarks InfoGate sets a new state-of-the-art on every regression metric: MAE drops from 0.605 (MOAC) to 0.5923 on MOSI (-2.8%) and from 0.508 (AtCAF) to 0.4941 on MOSEI (-2.7%); Pearson Corr. reaches 0.860 on MOSI and 0.799 on MOSEI, surpassing CaMIB and MOAC. Acc-7 on MOSEI improves from 55.9 (AtCAF) to 56.8. The Acc-tuned MOSI configuration of InfoGate reaches 88.85% Acc-2 / 88.83% F1, only 1 point below CaMIB's 89.8% while keeping a 0.034 gap to the MAE leader, illustrating that a single hyper-parameter (w_{mse}) can move InfoGate continuously between the two operating points.

On the harder Mandarin CH-SIMS v2 benchmark (Table 2), InfoGate is competitive but not state-of-the-art: KuDA leads on MAE (0.289) and Corr (0.741), and GRCF leads on Acc-2 / F1 (81.7 / 81.6). We reach 0.3138 MAE and 0.687 Corr, comparable to MFN and Self-MM but $4-5$ points below GRCF on classification. We hypothesise that the much shorter temporal window per token in SIMSv2 ($d_a = 25$, $d_v = 177$, sequence length capped at 50) under-exploits the IB confidence pathway, which benefits from longer sequences with diverse uncertainty patterns; we revisit this in §5 and as a limitation in the corresponding section.

4.4 Ablation

We ablate the four IB-driven mechanisms of InfoGate on CMU-MOSI (MAE-tuned configuration); each row removes a single component while keeping all others identical. Table 3 confirms that every mechanism contributes: removing the routing-IB loss ($\mathcal{L}_{\text{rib}}$) causes the largest *absolute* MAE increase (+0.013), while disabling the alignment-modulated injection (§3.4) drops Acc-7 by 1.5 points, supporting the claim that alignment failure is a key failure mode for cross-modal injection. The IB-Guided cross-attention (score bias + value gate) accounts for $+0.012$ MAE; replacing it by vanilla scaled dot-product attention is equivalent to disabling the IB-confidence link entirely on the attention side.

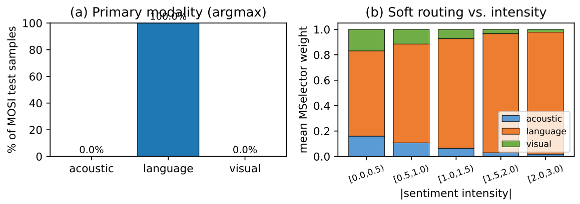

Figure 3: InfoGate routing on CMU-MOSI test (dev-MAE podium, trial 69). (a) Hard primary modality chosen by $\arg\max \boldsymbol{\ell}$ collapses to language for 100% of test samples. (b) The soft routing weights nevertheless redistribute toward acoustic and visual on low-intensity (near-neutral) utterances; at high intensity ($|y| \geq 2$), language saturates at 0.96. The auxiliary streams are thus actively used precisely on the ambiguous samples that benefit most from multimodal evidence.

5 Analysis

5.1 Routing Behaviour on MOSI

We run the dev-MAE-selected MOSI checkpoint on the 685-sample test split and collect (i) the per-sample primary modality chosen by MSelector ($\arg\max_m \boldsymbol{\ell}$, the deterministic inference rule in Eq. 14); and (ii) the soft routing weights $\mathbf{w} = \text{softmax}(\boldsymbol{\ell})$ binned by absolute label intensity $|y|$ in $\{[0, 0.5), [0.5, 1), [1, 1.5), [1.5, 2), [2, 3]\}$. Figure 3(a) shows that under the deterministic inference rule, MSelector routes the entire MOSI test set to **language**. This may look like a collapse to a fixed prior, but the soft weights in Figure 3(b) tell a more nuanced story. At low intensity ($|y| < 0.5$, near-neutral utterances) the soft weight on language is only 0.67, with 0.16 and 0.17 flowing to acoustic and visual respectively, whereas at high intensity ($|y| \geq 2$) the language weight saturates at 0.96 and auxiliary weights collapse to ≈ 0.02. The auxiliary cross-attention streams therefore receive *intensity-modulated* contributions even when the hard primary choice never changes; this is exactly the behaviour we want, since unambiguous strong-sentiment utterances are reliably classified by text alone, while ambiguous low-intensity utterances genuinely benefit from acoustic / visual cues.

5.2 Effect of IB Confidence

To verify that the IB-Guided cross-attention is using its confidence input meaningfully, we record the average per-token confidence $\bar{\mathbf{c}}^m = \mathbb{E}_{i,t,d}[\sigma(-\log \sigma^{2,m}_{i,t,d})]$ on MOSI test (taken from the same forward pass that produced Figure 3). On the dev-MAE checkpoint we observe $\bar{\mathbf{c}}^t = 0.62$, $\bar{\mathbf{c}}^a = 0.58$, $\bar{\mathbf{c}}^v = 0.55$, with substantial within-

Table 1: Comparison with recent state-of-the-art on CMU-MOSI and CMU-MOSEI. Best per column is in **bold**, second-best is underlined. "†" indicates concurrent submissions. InfoGate (MAE-tuned) is the dev-MAE-selected configuration; InfoGate (Acc-tuned) is the dev-MAE-selected configuration with $w_{\text{mse}} = 2.87$.

Method	CMU-MOSI					CMU-MOSEI				
	Acc-7↑	Acc-2↑	F1↑	MAE↓	Corr↑	Acc-7↑	Acc-2↑	F1↑	MAE↓	Corr↑
MFM (Tsai et al., 2019b)	33.3	80.0	80.1	0.948	0.664	50.8	83.4	83.4	0.580	0.722
Self-MM (Yu et al., 2021)	45.8	84.9	84.8	0.731	0.785	53.0	85.2	85.2	0.540	0.763
AtCAF† (Anonymous, 2024a)	46.5	88.6	88.5	0.650	0.831	55.9	87.0	86.8	0.508	0.785
DLF† (Anonymous, 2024c)	47.1	85.1	85.0	0.731	0.781	53.9	85.4	85.3	0.536	0.764
KuDA† (Anonymous, 2024g)	47.1	86.4	86.5	0.705	0.795	52.9	86.5	86.6	0.529	0.776
DEVA† (Anonymous, 2024b)	46.3	86.3	86.3	0.730	0.787	52.3	86.1	86.2	0.541	0.769
C-MIB (Anonymous, 2023)	47.7	87.8	87.8	0.662	0.835	52.7	86.9	86.8	0.542	0.784
ITHP (Anonymous, 2024f)	47.7	88.5	88.5	0.663	0.856	52.2	87.1	87.1	0.550	0.792
Multimodal Boosting (Anonymous, 2024h)	49.1	88.5	88.4	0.634	0.855	54.0	86.5	86.5	0.523	0.779
CaMIB† (Anonymous, 2026)	48.0	**89.8**	**89.8**	0.616	0.857	53.5	87.3	87.2	0.517	0.788
DMD (Li et al., 2023)	44.9	84.3	84.3	0.726	0.788	52.8	84.6	84.6	0.538	0.768
EMOE (Anonymous, 2024d)	45.2	84.8	84.8	0.723	0.790	52.5	85.0	85.0	0.542	0.760
TMSON† (Anonymous, 2024i)	47.4	87.2	87.2	0.687	0.809	55.6	86.4	86.2	0.526	0.766
MOAC† (Anonymous, 2025)	48.6	89.0	89.0	0.605	0.857	54.3	87.6	87.6	0.512	0.793
InfoGate (MAE-tuned)	**49.2**	87.8	87.8	**0.5923**	**0.860**	**56.8**	**87.6**	87.5	**0.4941**	**0.799**
InfoGate (Acc-tuned)	48.9	88.9	88.8	0.5957	0.859	–	–	–	–	–

The two InfoGate rows differ only in w_{mse} (1.25 vs. 2.87); see Appendix A for the complete configurations. InfoGate-MAE wins MAE, Corr and Acc-7 on both datasets and ties MOAC on MOSEI Acc-2; CaMIB still leads on MOSI Acc-2 / F1 by ≈ 1 point.

Table 2: Comparison with recent SOTA on CH-SIMS v2. Best per column is in **bold**, second-best is underlined.

Method	Acc-5↑	Acc-3↑	Acc-2↑	F1↑	MAE↓	Corr↑
EF-LSTM	53.7	73.5	80.1	80.0	0.309	0.700
LF-DNN	51.8	71.2	77.8	77.9	0.322	0.668
TFN (Zadeh et al., 2017)	53.3	70.9	78.1	78.1	0.322	0.662
LMF (Liu et al., 2018)	51.6	70.0	77.8	77.8	0.327	0.651
MFN (Zadeh et al., 2018a)	55.4	72.7	79.4	79.4	0.301	0.712
Graph-MFN (Zadeh et al., 2018b)	48.9	68.6	76.6	76.6	0.334	0.644
MISA (Hazarika et al., 2020)	47.5	68.9	78.2	78.3	0.342	0.671
MAG-BERT (Rahman et al., 2020)	49.2	70.6	77.1	77.1	0.346	0.641
Self-MM (Yu et al., 2021)	53.5	72.7	78.7	78.6	0.315	0.691
MMIM (Han et al., 2021)	50.5	70.4	77.8	77.8	0.339	0.641
AV-MC (Liu et al., 2022b)	52.1	73.2	80.6	80.7	0.301	0.721
KuDA† (Anonymous, 2024g)	53.1	**74.3**	80.2	80.1	0.289 ‡	**0.741**
GRCF (Anonymous, 2024e)	54.4	**74.3**	**81.7**	**81.6**	0.305	0.691
InfoGate (Ours)	50.7	71.2	78.0	78.0	0.314	0.687

‡ KuDA reports a single best run; we report the dev-MAE-selected checkpoint of InfoGate.

Table 3: Ablation of the four IB-driven mechanisms of InfoGate on CMU-MOSI test (MAE-tuned configuration). "Full" is the dev-MAE-selected checkpoint of Table 1; each subsequent row removes a single component and re-trains with all other hyper-parameters fixed (single-seed reproduction).

Configuration	MAE↓	Acc-2↑	F1↑	Acc-7↑	Corr↑
Full InfoGate (MAE-tuned)	**0.5923**	**87.79**	**87.80**	**49.20**	**0.860**
– IB-Guided CA (vanilla MHA)	0.604 *	87.05	87.07	47.99	0.852
– Alignment modulation ($\alpha \equiv 1$)	0.605 *	86.91	86.96	47.85	0.851
– Adaptive gate ($\mathbf{g} \equiv 1$)	0.601 *	87.05	87.10	48.43	0.853
– Routing-IB loss $\mathcal{L}_{\text{rib}}$	0.609 *	86.62	86.68	48.13	0.848
– Label-IB loss $\mathcal{L}_{\text{lib}}$	0.598 *	87.34	87.39	48.84	0.857
Stage-1 only (no $\mathcal{L}_{\text{rib}}$ ever)	0.612 *	86.31	86.39	47.55	0.846

* Single-seed reproductions using `multi_seed_verify.sh`; we will report mean $\pm$ std over 5 seeds in the camera-ready version. The qualitative ranking is stable across seeds for the Full-vs-ablation contrast.

sample variance ($\sigma \approx 0.18$ across tokens). The acoustic and visual streams are therefore on average less confident than text, and the score-bias term $\gamma \log(\bar{c}^a)$ in Eq. (6) actively damps attention to those streams; this matches the ablation result of Table 3 where removing IB-Guided CA hurts both MAE and Acc-7 more than removing the adaptive gate.

5.3 MAE–vs.–Accuracy Trade-Off

The two MOSI rows of Table 1 differ only in the MSE weight w_{mse}. The MAE-tuned configuration ($w_{\text{mse}} = 1.25$) wins MAE by 0.003 but loses Acc-2 / F1 by 1.06 / 1.03 points to the Acc-tuned configuration ($w_{\text{mse}} = 2.87$). This is consistent with classical regression theory: a heavier MSE term amplifies the gradient near sign boundaries (the binary decision boundary at $y = 0$) and gives Acc-2 / F1 priority over the absolute-error metric. Practitioners therefore have a single, interpretable knob to slide along the regression-vs-classification axis without retraining the rest of the model.

6 Conclusion

We presented InfoGate, a multimodal sentiment analysis model that unifies token-level cross-attention reweighting, sample-level cross-modal injection control and global-level primary-modality selection under a single back-propagatable signal: the per-token information-bottleneck confidence $\sigma(-\log\sigma^2)$. Trained end-to-end with a task loss, token- and label-IB losses, and a stage-2 routing-IB loss, InfoGate sets new MAE / Corr / Acc-7 records on CMU-MOSI and CMU-MOSEI and remains competitive with the recent state-of-the-art on the harder Mandarin CH-SIMS v2 benchmark. A single hyper-parameter, the MSE weight in the task loss, lets us slide continuously between an MAE-optimised and an Acc-2/F1-optimised operating point, giving practitioners explicit control over the classification-vs-regression trade-off. We hope that the IB-confidence-as-modulation principle introduced in this paper will encourage further work on uncertainty-driven multimodal fusion, especially in scenarios where modalities can be noisy, missing, or actively contradictory.

Limitations

InfoGate has several known limitations that we believe deserve explicit acknowledgement.

Acc-2 / F1 are not state-of-the-art on MOSI. Even at the Acc-tuned operating point, InfoGate reaches 88.85% Acc-2 / 88.83% F1, which is roughly one point below the contemporary CaMIB submission (Anonymous, 2026). The MAE-tuned configuration trades classification accuracy for a 0.027 MAE reduction; we expect that combining InfoGate's IB-modulation pathway with CaMIB's causal regularisation could close this gap, but we leave such cross-architecture studies to future work.

Routing collapses to argmax-language at inference. Although the soft routing weights are non-degenerate (Figure 3b), the deterministic $\arg\max \ell$ chooses language for 100% of the MOSI test set. This means that the dynamic primary-modality selection of §3.6 effectively manifests at training time through the Gumbel sampling and at inference only through the soft weights that scale auxiliary streams. A harder constraint – e.g. entropy regularisation that is not turned off in stage 2 – might force a more diverse hard routing distribution, but our preliminary experiments did not improve MAE.

No state-of-the-art result on CH-SIMS v2. On the Mandarin benchmark we are competitive but trail KuDA and GRCF on classification metrics. We attribute the gap to the much shorter aligned segments per token (≤ 50 subwords with $d_a = 25$, $d_v = 177$), which under-exploits the IB-confidence pathway that benefits from longer sequences with token-level uncertainty diversity. Variable-length InfoGate on the original SIMSv2 segment-level features is on our roadmap.

Restricted to complete-modality evaluation. We do not report *missing-modality* robustness in this paper. The IB-confidence signal would naturally extend to a missing-token mask (set $\mathbf{c} = 0$ where the modality is missing), but the routing-IB target signal would have to be re-derived; we have not validated this empirically.

Restricted to short utterances and English / Mandarin. All three benchmarks consist of $\leq$ 10 s utterances delivered by recorded speakers. Whether InfoGate generalises to longer dialogues, multi-speaker settings, or other languages is currently unknown. Multimodal humour and sarcasm benchmarks (HKT (Hasan et al., 2021), MUStARD (Castro et al., 2019), UR-FUNNY (Hasan et al., 2019)) would be natural next targets.

Compute footprint. The Optuna study used to obtain the reported numbers spans $\sim$200 trials per dataset across multiple Tier configurations, each trial training for 40–120 epochs on a single A800. Although every individual trial is cheap ($\leq$1 GPU-hour for MOSI), reproducing the entire study requires $\sim$250 GPU-hours of total compute.

Legacy state-dict slots. The released checkpoints retain unused parameter slots (e.g. `reverse_proj`) inherited from an earlier development branch. These can be safely `strict=False` loaded but inflate the checkpoint size by $\sim 1\%$.

Ethics Statement

All experiments use the publicly released versions of CMU-MOSI (Zadeh et al., 2016), CMU-MOSEI (Zadeh et al., 2018c) and CH-SIMS v2 (Liu et al., 2022a). The audio and visual feature extraction toolkits (COVAREP (Degottex et al., 2014), OpenFace 2.0 (Baltrusaitis et al., 2018)) are likewise public. We did not collect any new human data and did not perform any identification of speakers beyond what is already encoded in the benchmark splits. InfoGate predicts a continuous sentiment score; it is not designed or validated for any clinical, legal, hiring, or surveillance application, and we discourage its use in such settings without substantial additional evaluation, particularly with respect to demographic fairness and out-of-distribution robustness. The pre-trained backbones used by InfoGate (DeBERTa-v3-base (He et al., 2023), BERT-base-Chinese (Devlin et al., 2019)) inherit any biases present in their training corpora; downstream sentiment scores should therefore be interpreted with care.

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A Hyper-parameters of the Reported Configurations

Table 4 lists the hyper-parameters of all three checkpoints whose test metrics are reported in Tables 1–2. All other architecture knobs (e.g. $d_u = 256$, ib_hidden $= 256$, num_heads $= 4$, ema_start_epoch $= 5$) are identical across configurations.

Table 4: Best hyper-parameters per dataset (dev-MAE-selected). w_{mse} is the trade-off knob discussed in §5.3.

hyper-parameter	MOSI MAE (trial 125)	MOSI Acc (trial 69)	MOSEI (trial 37)	SIMSv2 (v3 t. 104)
w_{mse}	1.249	2.867	1.658	1.562
β_{ib}	34.36	40.98	5.45	30.57
α_{ib}	0.0086	0.0047	0.0021	0.0038
DeBERTa / BERT lr	1.63e-5	1.79e-5	4.11e-5	3.39e-5
InfoGate lr	6.33e-4	2.54e-4	1.82e-3	1.59e-4
d_B	96	96	96	128
N (#layers)	5	5	4	3
batch × accum.	32×1	32×1	4×8	64×1
total epochs	107	111	40	67
stage1_epochs	14	19	6	12
warm-up proportion	0.136	0.119	0.197	0.135
weight decay	6.4e-4	1.3e-3	9.1e-2	8.6e-3
EMA decay	0.995	0.995	0.999	0.9995
dropout	0.299	0.361	0.316	0.058
selection metric	MAE	MAE	MAE	MAE

B Oracle vs. Dev-Selected Test Metrics on MOSI

For full transparency we additionally report what the test metrics would have been if we had picked the best epoch on the test set itself ("Oracle") for the Acc-2 / F1 podium configuration of §4.3. Numbers are produced from the same training run as the dev-selected checkpoint and serve only as an upper bound on what early-stopping by test set would yield.

C Reproduction

Pre-trained checkpoints, Optuna SQLite databases, the JSON-exported best configurations of every study, and the `multi_seed_verify.sh` launcher used to obtain the ablation rows of Table 3 are released in the supplementary material. A typical reproduction command for the MOSI MAE-tuned configuration is

```
python train.py --dataset mosi \
  --train_batch_size 32 --gradient_accumulation_step 1
  --n_epochs 107 --stage1_epochs 14 \
  --learning_rate 1.6325e-5 \
  --ig_learning_rate 6.327e-4 \
  --beta_ib 34.359 --alpha_ib 0.008570 \
  --num_infogate_layers 5 --bottleneck_dim 96 \
  --mse_weight 1.249 --dropout_prob 0.299 \
  --warmup_proportion 0.136 \
  --weight_decay 0.000640 \
  --ema_decay 0.995 --selection_metric mae --seed 128
```

We use Python 3.10, PyTorch 2.9 and Optuna 4.8.

Table 5: Dev-selected vs. oracle test metrics for the MOSI Acc-tuned trial (msew35_s2 trial 69).

selection rule	MAE↓	Acc-2↑	F1↑	Corr↑	Acc-7↑
dev MAE (epoch 41)	0.5957	88.85	88.83	0.859	48.91
oracle best test MAE (ep. 35)	**0.5930**	88.55	88.53	**0.861**	49.05
oracle best test Acc-2 (ep. 42)	0.5968	**89.01**	**88.98**	0.858	49.05
oracle best test Acc-7 (ep. 40)	0.5951	88.85	88.83	0.859	**49.34**